

Element D

Design Concept Generation, Analysis, And Selection

The design the team has decided to build is a stair like structure with three rows at different elevations to properly solve all the problems listed. The light that is a part of the structure will be adjustable to go along with the height of the plant. There are other features along with the light that are considerate to the plants growth and future vegetation. In the beginning when the team was separated into groups, each group created a [prototype with different features and designs](#). Using the process of elimination to rule out insufficient features and deciding on a prototype that included attainable but successful components that could be appropriately created in a mindful time and budget. The team decided on a final design that is pictured in Element D without the light and pump included. The team has decided on the staircase prototype due to the fact that this design was easier to build and less complicated than the rest, which is what our team needed because of the time frame given to provide the Biology teacher with the unit. Not only that but this design is able to accomplish what the other designs had with less space usage, the uniqueness, the high number of alterable materials, and its different use of the pump in this system. Our design team gathered together and pointed out flaws and wanted aspects of three designs, eliminating ones that didn't seem

to achieve our goal or ones that seemed unattainable in our time frame. The team has also altered our final design implementing aspects from other designs to get our wanted final design. So far we've created a list of materials to use for our hydroponic unit. We've conducted research on the specific items needed to enact our solution, their prices, and exact measurements as well as some alternatives to those items if their cost exceeds our budget. As a team we also created a 3D model and hand-sketched model of our unit to give ourselves a reference of what our stakeholders want it to look like when our team will bring it to life. To ensure others will be able to replicate the results, all steps will be documented in order for it to be easy to follow. Additionally, pictures, our brainstorming process, and any research our team did will be documented as well, accessible to anyone who would like to replicate our results. The documentation of any challenges or issues the team has run into will also be documented and shared with the public, so that anyone replicating our results will not have to deal with problems that our team faced.

